DEFENCE JOURNEY

NDA Mock Test 2

Practice test in the NDA pattern with answer key and explanations

NDA | MOCK 2

TIP: Time: 40 minutes. Questions: 30 (Maths 12, English 8, General Knowledge 10). Marking: +1 for a correct answer, −1/3 for a wrong answer, 0 if left blank. This is a short practice test in the NDA pattern, not the full-length paper. Attempt it without looking at the notes, then check the answer key on the last page.

Part A: Mathematics

Q1. The number of subsets of a set having 3 elements is

(A) 8
(B) 6
(C) 9
(D) 3

Q2. The value of $(1 + i)^2$ is

(A) 2
(B) −2i
(C) 2i
(D) 1 + 2i

Q3. The sum of the first 20 natural numbers is

(A) 210
(B) 190
(C) 200
(D) 420

Q4. The discriminant of $2x^2 - 4x + 2 = 0$ is

(A) 8
(B) −8
(C) 0
(D) 16

Q5. The value of $\log_{10} 1000$ is

(A) 2
(B) 3
(C) 10
(D) 100

Q6. The number of ways of arranging the letters of the word MATH is

(A) 24
(B) 16
(C) 12
(D) 8

Q7. If sin A = 1/2, then cos 2A is

(A) √3/2
(B) 0
(C) 1
(D) 1/2

Q8. The slope of the line $3x + 2y = 6$ is

(A) $-3/2$
(B) $-2/3$
(C) $3/2$
(D) $2/3$

Q9. The radius of the circle $x^2 + y^2 - 6x + 8y = 0$ is

(A) 4
(B) 5
(C) 3
(D) 25

Q10. The derivative of e^{2x} at $x = 0$ is

(A) 1
(B) e
(C) 2
(D) 2e

Q11. The value of $\int_0^{\pi/2} \cos x \, dx$ is

(A) 0
(B) 1/2
(C) 2
(D) 1

Q12. Two dice are thrown together. The probability that the sum is 7 is

(A) 7/36
(B) 1/9
(C) 1/6
(D) 1/12

Part B: English

Q13. Choose the word closest in meaning to DILIGENT.

(A) Hardworking
(B) Careless
(C) Slow
(D) Lazy

Q14. Choose the word opposite in meaning to FRUGAL.

(A) Careful
(B) Thrifty
(C) Extravagant
(D) Economical

Q15. Fill in the blank: The news ___ true.

(A) is
(B) have been
(C) are
(D) were

Q16. What does 'to burn the midnight oil' mean?

(A) To start a fire
(B) To work late at night
(C) To sleep early
(D) To waste money

Q17. One word for 'fear of closed spaces':

(A) Claustrophobia
(B) Hydrophobia
(C) Xenophobia
(D) Acrophobia

Q18. Fill in the blank: He insisted ___ paying the bill.

(A) at
(B) to
(C) on
(D) for

Q19. Choose the correct sentence.

(A) Each of the players were given a medal.
(B) Each of the player has received a medal.
(C) Each of the players have received a medal.
(D) Each of the players has received a medal.

Q20. Change to indirect speech: He said, "I am tired."

(A) He said that he was tired.
(B) He said that he is tired.
(C) He said that I was tired.
(D) He said that he had been tired.

Part C: General Knowledge

Q21. Which planet is known as the Red Planet?

(A) Jupiter
(B) Mercury
(C) Venus
(D) Mars

Q22. Who founded the Maurya dynasty?

(A) Chandragupta Maurya
(B) Ashoka
(C) Samudragupta
(D) Bindusara

Q23. Which gland is called the 'master gland' of the body?

(A) Thyroid
(B) Pituitary
(C) Pancreas
(D) Adrenal

Q24. The Dandi March of Mahatma Gandhi took place in

(A) 1930
(B) 1942
(C) 1922
(D) 1919

Q25. The hardest natural substance is

(A) Quartz
(B) Diamond
(C) Iron
(D) Graphite

Q26. The National Defence Academy is located at

(A) Chennai
(B) Dehradun
(C) Hyderabad
(D) Khadakwasla, Pune

Q27. Which river is known as 'Dakshin Ganga'?

(A) Kaveri
(B) Godavari
(C) Krishna
(D) Mahanadi

Q28. Which vitamin is made in the skin in the presence of sunlight?

(A) Vitamin C
(B) Vitamin B_{12}
(C) Vitamin A
(D) Vitamin D

Q29. The highest mountain peak in the world is

(A) Nanda Devi
(B) Mount Everest
(C) Kanchenjunga
(D) K2

Q30. Who was the last Viceroy of India?

(A) Lord Curzon
(B) Lord Canning
(C) Lord Wavell
(D) Lord Mountbatten

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	A	2	C	3	A	4	C	5	B
6	A	7	D	8	A	9	B	10	C
11	D	12	C	13	A	14	C	15	A
16	B	17	A	18	C	19	D	20	A
21	D	22	A	23	B	24	A	25	B
26	D	27	B	28	D	29	B	30	D

Explanations

Q1. (A) A set with n elements has 2^n subsets: $2^3 = 8$.

Q2. (C) $(1 + i)^2 = 1 + 2i + i^2 = 1 + 2i - 1 = 2i$.

Q3. (A) $n(n + 1)/2 = 20 \times 21/2 = 210$.

Q4. (C) $b^2 - 4ac = 16 - 4(2)(2) = 0$, so the roots are equal.

Q5. (B) $10^3 = 1000$.

Q6. (A) All 4 letters are different: $4! = 24$.

Q7. (D) $\cos 2A = 1 - 2\sin^2 A = 1 - 2(1/4) = 1/2$.

Q8. (A) $y = -(3/2)x + 3$, so the slope is $-3/2$.

Q9. (B) Centre $(3, -4)$. Radius $= \sqrt{(9 + 16 - 0)} = 5$.

Q10. (C) $d/dx\ e^{2x} = 2e^{2x}$. At $x = 0$ this is 2.

Q11. (D) $[\sin x]$ from 0 to $\pi/2 = 1 - 0 = 1$.

Q12. (C) Favourable outcomes: (1,6), (2,5), (3,4), (4,3), (5,2), (6,1) = 6. Probability = 6/36 = 1/6.

Q13. (A) A diligent person works with care and steady effort.

Q14. (C) Frugal means careful with money. Extravagant is the opposite.

Q15. (A) *News* is singular in form and takes a singular verb.

Q16. (B) It means to study or work until late at night.

Q17. (A) Acrophobia is fear of heights, hydrophobia of water, xenophobia of strangers.

Q18. (C) The verb *insist* takes the preposition *on*.

Q19. (D) *Each* is singular and takes *has*, and the noun after 'of' is plural (players).

Q20. (A) In reported speech, the present tense 'am' becomes the past 'was', and 'I' becomes 'he'.

Q21. (D) The iron oxide on its surface gives Mars its red colour.

Q22. (A) Chandragupta Maurya founded it in about 321 BCE with Chanakya's help.

Q23. (B) The pituitary controls many other endocrine glands.

Q24. (A) It began on 12 March 1930 as a protest against the salt law.

Q25. (B) Diamond is a form of carbon with a rigid crystal structure.

Q26. (D) NDA is at Khadakwasla near Pune. IMA is at Dehradun, OTA at Chennai.

Q27. (B) The Godavari is the longest peninsular river.

Q28. (D) Sunlight helps the skin make vitamin D, which prevents rickets.

Q29. (B) Mount Everest (8,849 m) on the Nepal-China border is the highest. K2 is second and Kanchenjunga is third.

Q30. (D) Mountbatten was the last Viceroy and the first Governor-General of free India.